Carlos Jurado

Academic Website — carlosjurado@utexas.edu

Education

PhD, Astronomy; The University of Texas at Austin 2024 -
Advisor: Keith Hawkins

MA, Astronomy; The University of Texas at Austin 2026
Thesis: Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk
Advisor: Keith Hawkins

BS, Astrophysics; University of California, Los Angeles (UCLA) 2024

Research

Graduate Student Researcher at UT Austin – Austin, TX Sept. 2024 – Present
Advisor: Keith Hawkins

Student Researcher at University of Notre Dame – Notre Dame, IN May 2023 – Aug. 2024
Advisor: Lauren Weiss

Combined radial velocity measurements of Kepler-34 and Kepler-35 from several spectrographs to place a strong mass upper limit on additional stellar and substellar companions in the two systems.

Student Researcher at UCLA – Los Angeles, CA May 2022 – June 2024
Advisor: Smadar Naoz

Utilized supercomputers to create population synthesis models and explore the spatial distribution of stellar remnants at the Galactic Center.

Publications

[1] **Jurado, C.**; Hawkins, K.; Hunt J. A. S.; Hackshaw, Z.; Filion C.; Frankel, N., Carr, C., "Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from kinematic ridges with Gaia", ApJ, 10.3847/1538-4357/ae346c

[2] **Jurado, C.**; Weiss, L. M.; Daclison L.; Tofflemire, B.M.; Orosz J. A.; Welsh, W. F., "Upper Limits on Stellar Companions to the Kepler-34 and Kepler-35 Systems", AJ, 10.3847/1538-3881/ada5f5

[3] **Jurado, C.**; Naoz, S.; Lam, C. Y.; Hoang, B.-M., "Natal Kicks from the Galactic Center and Implications on their Environment and the Roman Space Telescope," ApJ, doi.org/10.3847/1538-4357/ad55ee

Conference Presentations

Division on Dynamical Astronomy Annual Meeting, Chicago, IL June 2026
Contributing Talk: "The Dynamical Origins of Azimuthal Metallicity Variations"

APS Far West Section, San Diego, CA — Oct. 2023
Contributing Talk: "The implications of Exploding Stars at the Galactic Center"

Seminars & Informal Talks

Stars, Planets, and ISM Seminar, The University of Texas at Austin — April 8 2026
Talk: "Exploring the Chemo-Kinematic Structure of the Milky Way Disk"

Smadar Naoz Research Group Meeting, UCLA — Jan. 8 2026
Talk: "Dynamical Origins Of Azimuthal Metallicity Variations"

Stars, Planets, and ISM Seminar, The University of Texas at Austin — Jan. 29 2025
Talk: "The Effects of Natal Kicks in the Galactic Center"

AstroBash, UCLA Department of Physics and Astronomy — Oct. 18 2023

AstroBash, UCLA Department of Physics and Astronomy — Nov. 7 2022

Teaching

Graduate Teaching Assistant, Stellar Astronomy (AST 352K)	Aug. 2026 – Dec. 2026
Graduate Teaching Assistant, Search For Extraterrestrial Life (AST 309L)	Aug. 2025 – Dec. 2025
Guest Lecture, Introductory Astronomy (AST 307)	Mar. 2025
Graduate Teaching Assistant, Introductory Astronomy (AST 301)	Jan. 2025 – May 2025
Undergraduate Course Reader, Introductory Physics (Physics 5A)	Oct. 2023 – Dec. 2023
Undergraduate Course Reader, Introductory Mechanics (Physics 1A)	April 2024 – June 2024
Undergraduate Learning Assistant, Introductory Mechanics (Physics 1A)	Jan. 2023 – Mar. 2023

Awarded Telescope Time

McDonald Observatory — Feb., July & Nov. 2026
PI: Carlos Jurado (15 nights)

Spectroscopically observe 10 stellar triples using the 2.7m Harlan J Smith Telescope in order to determine the level of chemical homogeneity.

Observing Experience

2.7m Harlan J. Smith Telescope, McDonald Observatory (21 nights)	Ft Davis, TX
0.8m Telescope, McDonald Observatory (2 nights)	Ft Davis, TX

Service & Outreach

Poster Presenter, Astronomy on Tap Austin 100th Show	July 2026
Graduate Student Assembly Representative, The University of Texas at Austin	2025 - 2026
Student Panelist, UCLA AstroLive Summer Workshop	July 2022
Volunteer and Student Panelist, UCLA Physics and Astronomy Outreach Event	May 2022

Skills

Programming Languages: Python, SQL/ADQL, C/C++
Tools: iSpec, SAPHIRES, galpy
Software: BACCHUS, IRAF, SAOImageDS9
Other: Linux/Unix, Git, LaTeX